


# HTML & CSS DEVELOPMENT LAB (BCS-554)

## B.Tech CSE 5th SEMESTER

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# Experiment 1: Starting a Git Repository

**1. Experiment Title:** Starting a Git Repository

**2. Aim:** Initialize a repository, configure identity, stage files and commit changes.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** Git tracks changes in a project. `git init` creates a repository, `git add` stages files and `git commit` records a snapshot.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  git init
02   git config --local user.name "Haseen Babu"
03   git config --local user.email "haseen@example.com"
04   git status
05   git add index.html
06   git commit -m "Add home page"
07   git log --oneline
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
$ git init
Initialized empty Git repository
$ git config user.name
Haseen Babu
$ git commit -m "Add home page"
[main (root-commit) a1b2c3d] Add home page
 1 file changed, 5 insertions(+)
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is Git?

2. What is the purpose of git init?

3. What is the difference between git add and git commit?

# Experiment 2: Pushing and Cloning a Repository

**1. Experiment Title:** Pushing and Cloning a Repository

**2. Aim:** Create a local bare remote, push commits and clone the repository.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** A remote repository stores shared project history. `git remote add` registers it, `git push` sends commits and `git clone` makes a working copy.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
git init --bare ../html-remote
 git remote add origin ../html-remote
 git remote -v
 git branch -M main
 git push -u origin main
 cd ..
 git clone html-remote html-copy
 cd html-copy
 git log --oneline
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
$ git remote -v
origin  ../html-remote (fetch)
origin  ../html-remote (push)
$ git push -u origin main
 * [new branch] main -> main
$ git clone html-remote html-copy
Cloning into html-copy...
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is a bare repository?

2. What does git push do?

3. What is the purpose of git clone?

# Experiment 3: Your First HTML Page

**1. Experiment Title:** Your First HTML Page

**2. Aim:** Create basic HTML pages using headings, paragraphs and text formatting.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** HTML provides the structure of a web page. The document includes html, head and body. Headings use h1–h6; paragraphs use p; strong and em add semantic emphasis.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<!DOCTYPE html>
<html lang="en">
<head><title>My First Page</title></head>
<body>
  <h1>Hello, I am Haseen.</h1>
  <h2>About HTML</h2>
  <h3>Learning Web Design</h3>
  <p>HTML structures web content.</p>
  <p>This is <strong>important</strong> and <em>useful</em>.<br>New line.</p>
</body>
</html>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
My First Page

Hello, I am Haseen.
About HTML
Learning Web Design
HTML structures web content.
This is important and useful.
New line.
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 4: Divs, Attributes and a Small Fix

**1. Experiment Title:** Divs, Attributes and a Small Fix

**2. Aim:** Group content with divs, use id and data attributes, and repair broken HTML.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** A div is a block-level generic container. The id attribute identifies an element, while data-* stores custom data. HTML tags should be properly nested and closed.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<div class="header">Website Header</div>
<div class="main">Main content</div>
<div class="footer">Website Footer</div>

<div id="profile">
  <p data-topic="hobbies">I enjoy reading.</p>
</div>

<h1>Attendance Report</h1>
<p>Asha: 92%</p>
<p>Rohan: 85%</p>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Website Header
Main content
Website Footer

I enjoy reading.

Attendance Report
Asha: 92%
Rohan: 85%
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?
2. Which HTML, CSS or Git feature is used here?
3. How can you verify that the output is correct?

# Experiment 5: Links and Images

**1. Experiment Title:** Links and Images

**2. Aim:** Link home and portfolio pages and display an image with alt text.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** Anchor elements use href to navigate. Images use src for the file path and alt for accessible alternative text.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  <h1>Welcome to My Site</h1>
02  <p>Visit my <a href="portfolio.html">portfolio</a>.</p>
03
04  <!-- portfolio.html -->
05  <h1>My Portfolio</h1>
06  <a href="home.html">Back to Home</a>
07
08  <!-- portfolio-image.html -->
09  <h1>My Picture</h1>
10  <img src="images/your-image.jpg" width="200" height="200"
11       alt="A description of the picture">
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Welcome to My Site
Visit my portfolio.  [click to open]

My Portfolio
Back to Home

My Picture
[ 200 × 200 image area ]
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 6: Types of HTML Lists

**1. Experiment Title:** Types of HTML Lists

**2. Aim:** Create unordered, nested ordered and description lists.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** ul creates bullets, ol creates numbered items, li defines list items, and dl/dt/dd create term-description pairs.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<h1>Shopping List</h1>
<ul><li>Notebook</li><li>Pen</li><li>Bag</li></ul>

<h1>Canteen Menu</h1>
<ol><li>Samosa</li><li>Sandwich
  <ol><li>Cheese</li><li>Vegetable</li></ol>
</li></ol>

<dl>
 <dt>How many books can I borrow?</dt>
 <dd>You can borrow five books.</dd>
 <dt>Is the library open on Sunday?</dt>
 <dd>No, it is closed.</dd>
</dl>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Shopping List
• Notebook
• Pen
• Bag

Canteen Menu
1. Samosa
2. Sandwich
   1. Cheese
   2. Vegetable

How many books can I borrow?
  You can borrow five books.
Is the library open on Sunday?
  No, it is closed.
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 7: About Me Page

**1. Experiment Title:** About Me Page

**2. Aim:** Build a semantic About Me page with navigation and a mailto contact link.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** header, nav, main, section and footer describe page regions. Fragment links navigate to ids; mailto opens an email application.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<header>
 <h1>Haseen Babu</h1>
 <nav><a href="#about">About</a>
      <a href="#contact">Contact</a></nav>
</header>
<main><section id="about">
 <h2>About me</h2>
 <p>I am a computer science student interested in web development.</p>
</section></main>
<footer id="contact">
 <p>Email: <a href="mailto:your-email@example.com">your-email@example.com</a></p>
</footer>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Haseen Babu
About   Contact

About me
I am a computer science student interested in web development.

Email: your-email@example.com
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 8: Sign-Up Forms and Choices

**1. Experiment Title:** Sign-Up Forms and Choices

**2. Aim:** Create labelled form fields, radio buttons, a dropdown and required validation.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** Forms collect user data. Labels connect to controls using for/id. Email and password inputs provide suitable input behaviour. required enables browser validation.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<form method="post" action="thanks.html">
 <label for="name">Full Name:</label>
 <input id="name" name="name" type="text" required>
 <label for="email">Email:</label>
 <input id="email" name="email" type="email" required>
 <label for="pass">Password:</label>
 <input id="pass" name="password" type="password" required>
 <p>Gender:</p>
 <input type="radio" name="gender" value="Male"> Male
 <input type="radio" name="gender" value="Female"> Female
 <select name="course"><option>BCA</option><option>B.Tech</option></select>
 <button type="submit">Sign Up</button>
</form>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Full Name: [_________________]
Email:      [_________________]
Password:   [***************]
Gender:     ( ) Male  ( ) Female
Course:     [BCA             v]
            [ Sign Up ]

Empty required field → browser validation message.
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 9: Tables with Headers and Footers

**1. Experiment Title:** Tables with Headers and Footers

**2. Aim:** Create a marks table with caption, table sections and accessible scope attributes.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** table organizes data in rows and columns. caption names the table; thead, tbody and tfoot group sections. th defines header cells and scope improves accessibility.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<table border="1">
 <caption>Marks of Class A</caption>
 <thead><tr><th scope="col">Name</th>
 <th scope="col">Maths</th><th scope="col">Science</th></tr></thead>
 <tbody>
  <tr><th scope="row">Asha</th><td>78</td><td>85</td></tr>
  <tr><th scope="row">Rohan</th><td>64</td><td>70</td></tr>
  <tr><th scope="row">Neha</th><td>91</td><td>88</td></tr>
 </tbody>
 <tfoot><tr><th>Class average</th><td>71</td><td>77.5</td></tr></tfoot>
</table>
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
            Marks of Class A
+-------+-------+---------+
| Name  | Maths | Science |
+-------+-------+---------+
| Asha  |  78   |   85    |
| Rohan |  64   |   70    |
| Neha  |  91   |   88    |
+-------+-------+---------+
| Average| 71   |  77.5   |
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 10: CSS Fundamentals

**1. Experiment Title:** CSS Fundamentals

**2. Aim:** Apply inline, internal and external CSS with element, class and id selectors.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** CSS styles HTML. Inline CSS uses style attributes, internal CSS uses a style block, and external CSS uses a separate stylesheet linked with link.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
<style>
 h1 { color: navy; }
 .highlight { background: yellow; }
 .note { font-style: italic; }
 #title { text-decoration: underline; }
 p { font-size: 18px; color: grey; }
</style>
<h1 id="title">CSS Fundamentals</h1>
<p style="color:red">This paragraph is red.</p>
<p class="highlight">Highlighted paragraph.</p>
<link rel="stylesheet" href="style.css">
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
CSS Fundamentals  (navy + underline)
This paragraph is red.
Highlighted paragraph (yellow background)

External CSS: note text italic; paragraphs grey, 18px.
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 11: Units and CSS Variables

**1. Experiment Title:** Units and CSS Variables

**2. Aim:** Use px, percentage and rem units and define reusable CSS custom properties.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** px is a fixed CSS length unit; % is relative to a containing block; rem is relative to root font size. Custom properties begin with -- and are read with var().

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
:root {
 --main-color: #1d3557;
 --accent-color: #e63946;
 --page-padding: 20px;
}
.fixed-box { width: 200px; height: 40px; background: lightblue; }
.half-box { width: 50%; height: 40px; background: lightgreen; }
body { color: var(--main-color); padding: var(--page-padding); }
h1 { color: var(--accent-color); }
p { font-size: 1.5rem; }
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
[ light blue box: 200px × 40px ]
[ light green box: 50% width × 40px ]

Paragraph text: 1.5rem
Heading: accent red
Body text: main navy
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 12: Box Model, Cards and Scrolling

**1. Experiment Title:** Box Model, Cards and Scrolling

**2. Aim:** Demonstrate box sizing, style a card and make a fixed-height box scrollable.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** The box model contains content, padding, border and margin. content-box adds padding and border outside the declared width; border-box includes them. overflow:auto enables scrolling.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  .box { width: 200px; padding: 20px;
02         border: 5px solid navy; margin: 15px; }
03  .content-box { box-sizing: content-box; }
04  .border-box { box-sizing: border-box; }
05  .card { background: #eaf4f4; border: 2px solid #1d3557;
06          border-radius: 12px; padding: 16px; }
07  .scroll-box { height: 120px; overflow: auto; }
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
content-box outer width = 250px
border-box outer width = 200px

+--------------------------+
| Welcome Card             |
| Rounded border + padding |
+--------------------------+

Scrollable area:
Line 1
Line 2
Line 3
[scroll to see remaining lines]
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 13: Flexbox Layouts

**1. Experiment Title:** Flexbox Layouts

**2. Aim:** Build a flex navigation bar, centred cards and wrapping stationery cards.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** Flexbox is a one-dimensional layout model. display:flex creates a flex container; justify-content and align-items align items; flex-wrap allows rows to wrap.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  nav { display: flex; gap: 24px; background: #222; padding: 16px; }
02  nav a { color: white; }
03  .row { display: flex; justify-content: center;
04         align-items: center; gap: 20px; }
05  .card { padding: 20px; width: 150px; }
06  .wrap { display: flex; flex-wrap: wrap; gap: 16px; }
07  .item { flex: 1 1 160px; }
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
[Home]   [About]   [Events]   [Contact]

[ Starter 199 ] [ Standard 349 ] [ Premium 499 ]

Stationery Shop
[Notebook 60] [Pen 15] [Bag 350]
[Book 120]
(cards wrap on narrow screens)
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 14: CSS Grid Layouts

**1. Experiment Title:** CSS Grid Layouts

**2. Aim:** Create equal columns, a responsive photo gallery and dashboard named areas.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** CSS Grid is a two-dimensional layout system. grid-template-columns defines columns, fr distributes available space, repeat/minmax support responsive tracks, and grid-template-areas names layout regions.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  .grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 12px; }
02  .gallery { display: grid;
03   grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));
04   gap: 10px; }
05  .dashboard { display: grid; grid-template-columns: 200px 1fr;
06   grid-template-areas: "header header" "sidebar main" "footer footer"; }
07  header { grid-area: header; } aside { grid-area: sidebar; }
08  main { grid-area: main; } footer { grid-area: footer; }
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Grid:
[ 1 ] [ 2 ] [ 3 ]
[ 4 ] [ 5 ] [ 6 ]

Club Photo Gallery:
[Photo 1] [Photo 2] [Photo 3]
[Photo 4] [Photo 5] [Photo 6]

[       Header       ]
[Sidebar] [Main      ]
[       Footer       ]
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?

# Experiment 15: Responsive Pages with Media Queries

**1. Experiment Title:** Responsive Pages with Media Queries

**2. Aim:** Use media queries for mobile layouts, responsive images and rem-based text.

**3. Software / Tools Required:** Git and terminal (Experiments 1–2); Visual Studio Code or any text editor; HTML5, CSS3 and Google Chrome / Firefox.

**4. Theory:** Responsive design adapts to viewport size. The viewport meta tag sets the layout viewport. Media queries apply styles at breakpoints; max-width:100% keeps images within containers.

**5. Algorithm / Procedure:** 1) Create the required file(s). 2) Enter the code shown below. 3) Save the file with the required name. 4) Run Git commands or open the HTML file in a browser. 5) Verify the output preview and capture your own screenshot.

**6. Program / Queries / Commands (editor-style code screenshot):**

VS CODE | HTML / CSS

```
01  .cards { display: flex; flex-direction: column; gap: 16px; }
02  @media (min-width: 600px) {
03   .cards { flex-direction: row; }
04   .card { flex: 1; }
05  }
06  img { max-width: 100%; height: auto; }
07  body { font-size: 1rem; }
08  h1 { font-size: 1.5rem; }
09  @media (min-width: 600px) { h1 { font-size: 2.5rem; } }
```

**7. Input:** The HTML/CSS source and sample values specified in the assignment; Git commands for Experiments 1–2.

**8. Output (browser / terminal screenshot-style preview):**

BROWSER / TERMINAL OUTPUT PREVIEW

```
Mobile (<600px):
[ Card One ]
[ Card Two ]

Desktop (≥600px):
[ Card One ] [ Card Two ]

Image scales to fit container.
Heading: 1.5rem mobile → 2.5rem desktop
```

**9. Result:** The experiment was completed successfully. The required HTML/CSS or Git operation was demonstrated and the expected result was checked.

**10. Viva-Voce Questions:**

1. What is the main purpose of this experiment?

2. Which HTML, CSS or Git feature is used here?

3. How can you verify that the output is correct?